

Experiment No: 04

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Title: Lists in Python

Program:

```
# List operations
```

```
lst = []
```

```
n = int(input("Enter number of elements: "))
```

```
for i in range(n):
```

```
    val = int(input("Enter element: "))
```

```
    lst.append(val)
```

```
print("\nOriginal List:", lst)
```

```
# Basic operations
```

```
lst.append(100)
```

```
print("After append:", lst)
```

```
lst.insert(1, 50)
```

```
print("After insert:", lst)
```

```
if len(lst) > 0:
```

```
    lst.pop()
```

```
print("After pop:", lst)
```

```
# Searching
```

```
search = int(input("Enter element to search: "))
```

```
if search in lst:
```

```
print("Element found at index:", lst.index(search))
```

else:

```
print("Element not found")
```

Sorting

```
print("Sorted List:", sorted(lst))
```

```
print("Maximum:", max(lst))
```

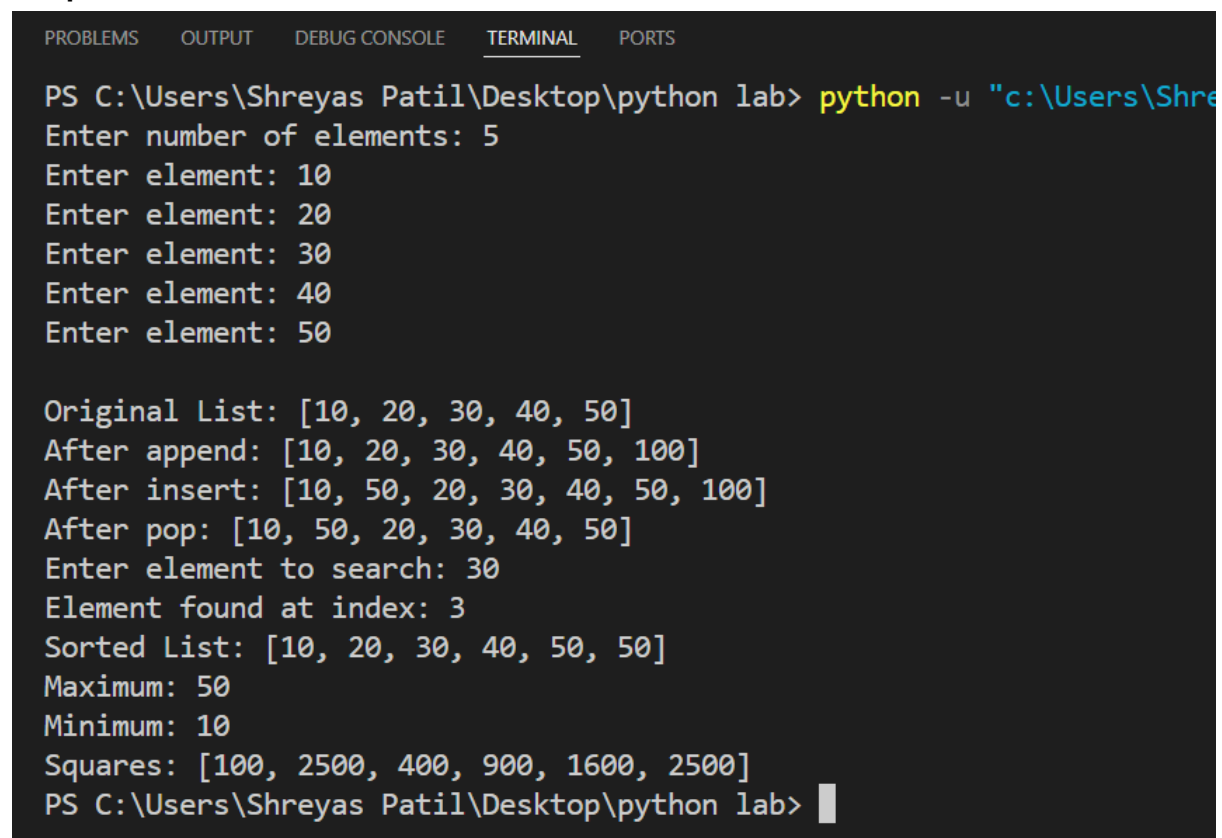
```
print("Minimum:", min(lst))
```

List comprehension

```
squares = [x**2 for x in lst]
```

```
print("Squares:", squares)
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Shreyas Patil\Desktop\python lab> python -u "c:\Users\Shre
Enter number of elements: 5
Enter element: 10
Enter element: 20
Enter element: 30
Enter element: 40
Enter element: 50

Original List: [10, 20, 30, 40, 50]
After append: [10, 20, 30, 40, 50, 100]
After insert: [10, 50, 20, 30, 40, 50, 100]
After pop: [10, 50, 20, 30, 40, 50]
Enter element to search: 30
Element found at index: 3
Sorted List: [10, 20, 30, 40, 50, 50]
Maximum: 50
Minimum: 10
Squares: [100, 2500, 400, 900, 1600, 2500]
PS C:\Users\Shreyas Patil\Desktop\python lab>
```